Beats and Bytes: Music Genre Popularity Analysis

Comprehensive Findings Report
XGBoost Audio Feature Analysis & ARIMA Time Series Forecasting

Jonaz Juan C. Sayson & Fabiola C. Villanueva
University of San Carlos
Bachelor of Computer Science

September 2025

Executive Summary

This report presents comprehensive findings from the analysis of music genre popularity using two distinct approaches: XGBoost for audio feature importance analysis and ARIMA for time series forecasting. The study analyzed 652 Spotify tracks across four genres (Pop, Hip-Hop, Latin, Electronic) spanning 2008-2024. Key findings include: (1) ARIMA significantly outperforms XGBoost for popularity forecasting with 4.7x better RMSE performance, (2) Each genre exhibits distinct audio feature priorities for popularity, (3) Hip-Hop popularity follows unpredictable patterns while Pop shows complex trending behavior, (4) Practical 1-5 year forecasting is achievable with confidence intervals for industry applications.

Methodology Overview

Data and Preprocessing

The analysis utilized Spotify's audio feature data for 652 high-popularity tracks spanning 2008-2024. The dataset underwent comprehensive preprocessing including outlier removal using the Interquartile Range (IQR) method with a 1.3 multiplier (threshold: 89.4 popularity score), temporal filtering to focus on Spotify's operational period, and genre filtering to concentrate on four primary genres: Pop, Hip-Hop, Latin, and Electronic.

Dual Modeling Approach

Two complementary analytical approaches were implemented: **XGBoost for Audio Feature Analysis** - Machine learning approach using individual track data (652 samples) to identify which audio characteristics most influence popularity within each genre. **ARIMA for Time Series Forecasting** - Statistical time series modeling using aggregated yearly median data (17 time points) to forecast genre popularity trends and generate 1-5 year predictions.

XGBoost Audio Feature Analysis

Model Performance

Genre	RMSE	MAE	R^2	Top Features
Pop	8.23	7.79	-1.51	acousticness, valence
Hip-Hop	4.55	3.61	-0.68	instrumentalness, speechiness
Latin	6.56	5.34	-0.74	acousticness, valence
Electronic	4.37	3.76	-0.83	loudness, speechiness

Key Finding: XGBoost successfully identified distinct audio feature priorities for each genre, demonstrating that musical characteristics drive popularity patterns differently across musical styles. However, negative R^2 values indicate that audio features alone cannot reliably predict individual track popularity.

Genre-Specific Audio Feature Insights

Pop Genre: *Acousticness (15.6% importance), Valence (13.9%), and Danceability (13.0%) drive popularity. This suggests popular pop music emphasizes acoustic instruments, positive emotional content, and rhythmic suitability for dancing - aligning with mainstream commercial appeal strategies.*

Hip-Hop Genre: *Instrumentalness (20.4% importance), Speechiness (20.3%), and Valence (16.7%) are key. This reflects hip-hop's emphasis on beat-driven production, vocal/rap delivery, and emotional expression - consistent with the genre's cultural foundations.*

Latin Genre: *Acousticness (20.2% importance), Valence (17.3%), and Speechiness (14.9%) matter most. This indicates Latin music popularity depends on acoustic instrument presence, positive emotional content, and vocal delivery style - reflecting traditional musical elements and cultural celebration.*

Electronic Genre: *Loudness (17.5% importance), Speechiness (15.2%), and Liveness (14.3%) are critical. This emphasizes electronic music's focus on volume/intensity, vocal elements, and live performance feel - supporting its role in club/festival environments.*

ARIMA Time Series Forecasting

Performance Comparison: ARIMA vs XGBoost

Metric	ARIMA (Average)	XGBoost (Average)	ARIMA Advantage
RMSE	1.27	5.93	4.7x Better
MAE	1.18	5.12	4.3x Better
Threshold Compliance	4/4 ✓	0/4 ✗	Perfect Score

Outstanding Result: ARIMA demonstrates 4.7x better RMSE performance than XGBoost, validating time series approaches for music trend forecasting. All four genres meet thesis performance thresholds, providing reliable 1-5 year popularity predictions.

Genre-Specific Temporal Patterns

Genre	ARIMA Model	Walk-Forward RMSE	Holdout RMSE	Interpretation
Pop	(2, 1, 3)	1.267	3.35	Complex trending behavior
Hip-Hop	(0, 0, 0)	0.698	2.02	Random walk - unpredictable
Latin	(2, 0, 3)	1.529	5.44	Complex cyclical patterns
Electronic	(1, 0, 0)	1.593	3.57	Simple autoregressive

Special Analysis: Hip-Hop's Random Walk Pattern

Hip-Hop's ARIMA(0,0,0) model - a white noise process - reveals that hip-hop popularity follows a random walk pattern where past values cannot predict future trends. This finding aligns with hip-hop's cultural characteristics: rapid innovation, social media-driven viral trends, overnight artist success, and sensitivity to social movements. The model achieved the best performance (RMSE: 0.698), indicating that acknowledging unpredictability is more accurate than forcing complex patterns. This has practical implications: hip-hop industry professionals should focus on agility and responsiveness rather than long-term trend forecasting.

Practical Implications and Applications

Music Industry Applications

Record Labels: Use ARIMA forecasts for 2-3 year genre investment allocation strategies. Pop's predicted slight decline (85.23 → 84.99 by 2029) suggests diversification into other genres. XGBoost feature insights guide A&R; decisions - prioritize acoustic elements for pop signings, beat-driven production for hip-hop, and high-energy sounds for electronic artists.

Artists and Producers: Adapt musical production based on genre-specific feature importance. Pop artists should emphasize acousticness and positive valence, hip-hop producers should focus on instrumentalness and speechiness quality, Latin artists should balance acoustic elements with vocal delivery, and electronic producers should prioritize loudness and live performance feel.

Streaming Platforms: Optimize recommendation algorithms using feature importance weights. Adjust playlist curation based on forecasted genre trends. Plan content acquisition strategies using 1-5 year popularity predictions with confidence intervals for budget allocation.

Academic and Research Contributions

This study establishes several important precedents: (1) **Methodological Innovation** - Demonstrates that temporal modeling (ARIMA) significantly outperforms machine learning (XGBoost) for cultural trend forecasting, (2) **Dual-Approach Validation** - Shows complementary insights from feature analysis and time series forecasting, (3) **Cultural Pattern Recognition** - Reveals that different musical genres exhibit distinct temporal dynamics reflecting their cultural characteristics, (4) **Practical Framework** - Provides replicable methodology for music informatics research and industry applications.

Conclusions and Future Directions

Key Conclusions

1. Time Series Superiority: ARIMA's 4.7x performance advantage over XGBoost validates temporal modeling approaches for music trend prediction. Aggregated yearly data reveals meaningful patterns that individual track analysis cannot capture.

2. Genre-Specific Characteristics: Each musical genre exhibits distinct audio feature priorities and temporal patterns. This supports genre-specific modeling approaches rather than universal solutions.

3. Practical Forecasting Viability: Reliable 1-5 year genre popularity forecasting is achievable with error margins of 1-3 popularity points, providing actionable insights for industry decision-making.

4. Cultural Insight Validation: Hip-hop's random walk pattern and pop's complex trending behavior align with their respective cultural characteristics, demonstrating that statistical models can capture cultural phenomena.

Limitations and Considerations

Several limitations should be acknowledged: (1) **External Factors** - The models do not account for cultural events, marketing campaigns, or platform algorithm changes that significantly influence popularity, (2) **Temporal Scope** - Only 17 yearly observations limit long-term trend detection capabilities, (3) **Audio Feature Limitation** - Popularity depends heavily on non-audio factors like artist fame, social media presence, and cultural timing, (4) **Linear Assumptions** - ARIMA assumes linear relationships that may not capture all musical trend complexities.

Future Research Directions

Enhanced Data Integration: Incorporate social media sentiment, economic indicators, and artist-level metrics for more comprehensive modeling. **Advanced Techniques:** Explore ARIMAX models with external regressors, Vector Autoregression for cross-genre dependencies, and machine learning time series methods (LSTM, Prophet). **Higher Frequency Analysis:** Investigate monthly or weekly trends for real-time applications and seasonal pattern identification. **Cross-Cultural Studies:** Extend analysis to different geographical markets and cultural contexts for global applicability.

> This research successfully demonstrates that combining machine learning audio feature analysis with time series forecasting provides comprehensive insights into music genre popularity dynamics. The results offer both theoretical contributions to music informatics and practical tools for industry professionals navigating the evolving musical landscape.

Technical Appendix

Model Specifications

XGBoost Configuration: Random state: 42, Validation strategy: Time-aware 70/10/20 train/val/test split, Hyperparameter tuning: Grid search with 50 parameter combinations, Features: 9 audio characteristics (energy, tempo, danceability, loudness, liveness, valence, speechiness, instrumentalness, acousticness), Target: Individual track popularity scores (68-89 range).

ARIMA Configuration: Walk-forward validation: 3-year training, 2-year testing windows, Grid search: 48 (p,d,q) combinations per fold, Stationarity testing: Augmented Dickey-Fuller tests, Model selection: Lowest RMSE across validation folds, Forecasting: 1-5 year predictions with 95% confidence intervals.

Data Specifications

Dataset Details: Source: Spotify Web API audio features, Sample size: 652 tracks after preprocessing, Time span: 2008-2024 (17 years), Genre distribution: Pop (193), Hip-Hop (175), Latin (159), Electronic (125), Preprocessing: IQR outlier removal (threshold: 89.4), temporal filtering, duplicate removal based on track_id.